



# FLUTTER MOBILE ENGINEERING

Technical Assessment 2025

**Objective:** Evaluate the candidate's competency in architecting production-grade, highly performant cross-platform mobile applications using modern Flutter framework conventions and clean engineering patterns.

## Project Brief

Develop an enterprise-grade mobile application module for booking and orchestrating home-care nursing services. The application must accurately simulate an authentic, real-world digital health workflow.

## Core & Extended Requirements

- **Identity & Session Management (Authentication):** Implement a cryptographically secure registration and login mechanism. Ensure secure session persistence.
- **Deterministic Booking System:** Users must seamlessly select specific clinical services, dynamically resolve schedule availabilities, input clinical patient remarks, and move through a deterministic state machine representing the booking.
- **Data-Driven User Dashboard:** Construct a responsive dashboard reflecting aggregate user states, active medical requests, and historical records.

## Architectural Expectations

- **Clean Architecture:** Strict structural segregation into independent layers (Presentation, Domain, Data).
- **State Management:** Limited to BLoC or Riverpod. Explicit separation of states (Initial, Loading, Success, Error).
- **Network Layer:** A unified HTTP client layer utilizing advanced network abstraction, HTTP Interceptors, and automated retry mechanisms.

## Technology Stack & Bonus

Flutter 3.x

Dart

flutter\_bloc

dio

freezed

Offline-First (Hive/Isar)

Firebase Push

CI/CD Pipeline



## CORPORATE EVALUATION PHILOSOPHY

### 1. ENGINEERING MINDSET

Your technical choices must be deliberate and justified. Design architectures to treat components as interchangeable blocks, backed by comprehensive documentation.

### 2. PRODUCTION READINESS

We evaluate code based on fitness for production. This requires thorough input validation, robust exception handling, and structural security.

### 3. ATTENTION TO DETAIL

This includes clean variable naming conventions, consistent formatting, comprehensive error messaging, and intuitive edge-case handling.

### 4. CODE DELIVERY PROTOCOL

We look for professional utilization of version control, semantic commit structures, and a polished presentation formatting layout for all deliverables.



### Service Level Agreement (SLA)

All core code, architecture blueprints, live links, and descriptive video demonstrations must be submitted within **72 Hours** from assignment delivery.